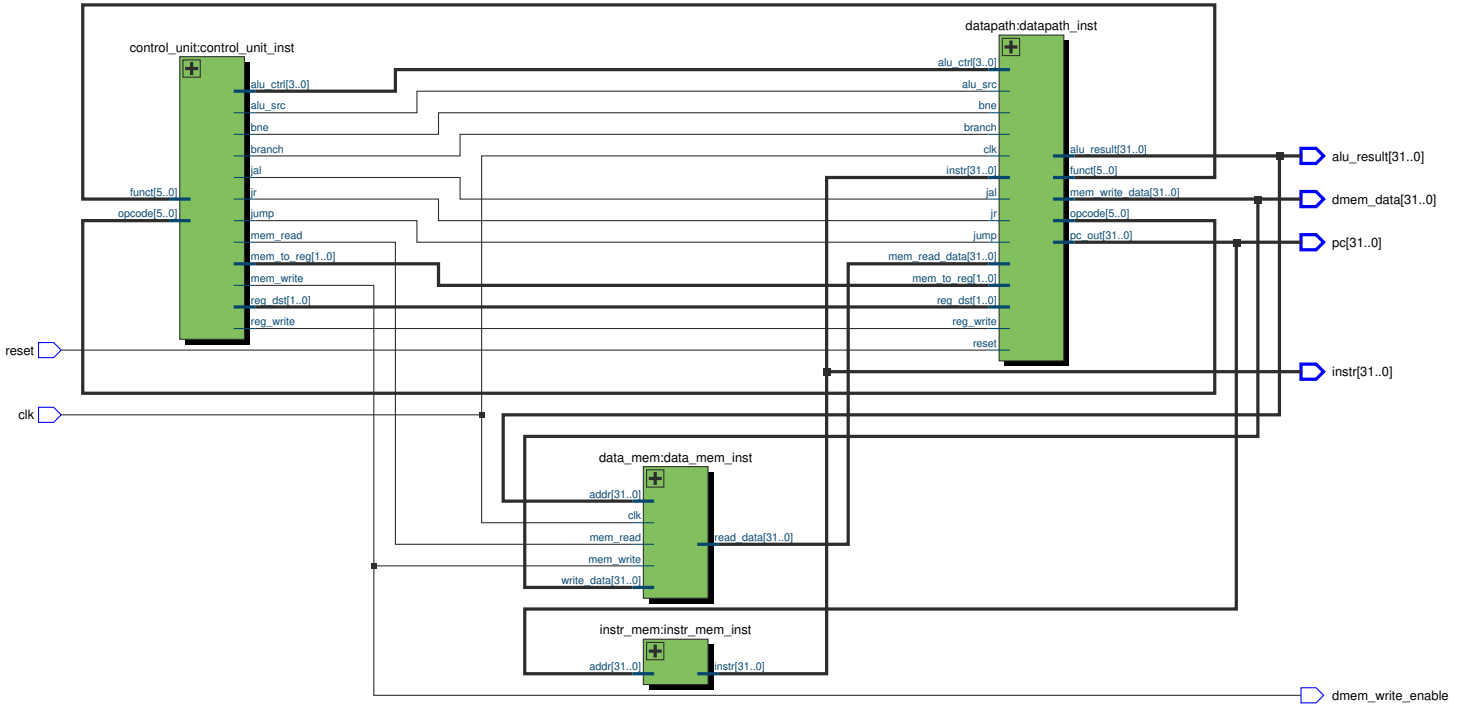
control_unit:control_unit_inst
alu_ctrl[3..0]
alu_src
bne
branch
jal
jr
jump
mem_read
mem_to_reg[1..0]
mem_write
reg_dst[1..0]
reg_write
funct[5..0]
opcode[5..0]
datapath:datapath_inst
alu_ctrl[3..0]
alu_src
bne
branch
clk
instr[31..0]
jal
jr
jump
mem_read_data[31..0]
mem_to_reg[1..0]
reg_dst[1..0]
reg_write
reset
alu_result[31..0]
funct[5..0]
mem_write_data[31..0]
opcode[5..0]
pc_out[31..0]
reset
clk
data_mem:data_mem_inst
addr[31..0]
clk
mem_read
mem_write
write_data[31..0]
read_data[31..0]
instr_mem:instr_mem_inst
addr[31..0]
instr[31..0]
alu_result[31..0]
dmem_data[31..0]
pc[31..0]
instr[31..0]
dmem_write_enable